

Project Title: "When an illness isn't alone: How can we improve insurance risk-pricing strategies through comorbidity-informed ICU survival prediction?"

Unit: ADS1002

Group: MY-G1-ICU

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1. Executive Summary

This project investigates how kidney dysfunction and unstable blood pressure influence ICU survival outcomes, with the goal of applying these findings to improve data-driven insurance risk pricing. Using a pre-processed ICU dataset containing demographic, physiological, and comorbidity-related variables, the analysis combines statistical exploration with predictive modelling to identify which clinical indicators most strongly determine survival probability.

The dataset was cleaned and refined through targeted imputation for missing values in SAPS.I (Simplified Acute Physiology Score), SOFA (Sequential Organ Failure Assessment), and gender variables. Exploratory data analysis revealed that patients with either kidney dysfunction or unstable blood pressure showed significantly higher mortality rates compared to those without these conditions. The mortality effect was even stronger among patients who exhibited both comorbidities at the same time.

To quantify predictive relationships, our study implemented supervised machine-learning models including Logistic Regression and Random Forest. Among them, the Random Forest classifier demonstrated the highest overall performance across accuracy, precision, recall, and AUC metrics. Feature importance analysis revealed that SOFA, SAPS.I, and the presence of kidney dysfunction were dominant predictors of mortality risk, followed closely by unstable blood pressure indicators.

Findings highlight the clinical and actuarial importance of comorbidity-aware prediction systems. By integrating medical indicators such as kidney dysfunction and blood pressure instability into predictive models, insurance systems can enhance fairness and accuracy in premium risk stratification.

2. Background and Project Description

Mortality in the ICU is heavily influenced by the interaction between acute physiological stress and underlying comorbidities. Two of the most significant risk factors are kidney dysfunction and unstable blood pressure, both of which directly affect organ failure progression and treatment complexity. For insurers, these comorbidities result in substantial differences in financial risk because they influence hospital stay duration, treatment intensity, and overall cost of care.

The dataset used in this study, titled “Preprocessed ICU Data”, contains 1,474 anonymised patient records and 232 monitored variables from ICU admissions. It includes demographic details such as age and gender, standardized ICU scoring measures such as SOFA and SAPS.I, vital signs (such as systolic and diastolic blood pressure, heart rate), and lab markers (e.g., blood urea nitrogen, creatinine, troponin, lactate). The outcome we're looking at, the targeted variable “In-hospital_death”, indicates whether a patient survived their hospital stay, enabling supervised learning methods to assess risk factors and predict mortality.

The dataset reflects the multidimensional and clinically complex nature of ICU patients, who often present with overlapping conditions such as cardiovascular disease and renal impairment. This makes it particularly suitable for examining how specific comorbidities, such as kidney dysfunction and blood pressure instability, affect survival probability and for developing predictive models that better represent real-world patterns of clinical risk.

For this project, the dataset was analysed in a Jupyter Notebook environment using Google Colab. The workflow involved several key steps: data cleaning, imputation of missing values, feature selection, exploratory visualisation, and predictive model development. Exploratory data analysis was performed to uncover patterns in mortality across different demographic, clinical, and comorbidity dimensions, providing the foundation for subsequent modelling and interpretation.

3. Pre-processing and Manipulation in Python

Once we loaded the data into Python, we refined it further and handled missing values using Pandas and Scikit-learn to ensure our analysis was accurate. We used several imputation techniques to address missing values and outliers.

First, we corrected the Gender variable using SimpleImputer with the "most frequent" strategy, replacing any incorrect or missing entries with the most common gender. For the SAPS.I and SOFA scores, which are critical indicators of illness severity, we imputed missing or invalid values (marked as "-1") using KNNImputer (with k=2). This method used the relationships between correlated features to produce realistic and consistent imputed values.

All continuous variables were normalized before imputation to avoid scaling issues. After that, we ran checks to confirm that no invalid or missing entries remained, ensuring the data was complete for subsequent analysis.

We also selected specific subsets of features for different analyses. For instance, we grouped cardiovascular variables (like systolic, diastolic, and mean arterial pressures) for blood pressure analysis, and isolated renal markers (such as urine output, creatinine, BUN, and lactate) for kidney function analysis. Feature engineering involved creating binary indicators for high blood pressure and calculating mortality-related ratios.

These steps helped turn the data into a strong setup, ready for visualizing, statistical analysis, and machine learning. This preparation phase sets the stage for exploratory data analysis, comorbidity-based modeling, and ultimately, for developing risk level grouped mortality predictions.

4. Exploratory Data Analysis (EDA)

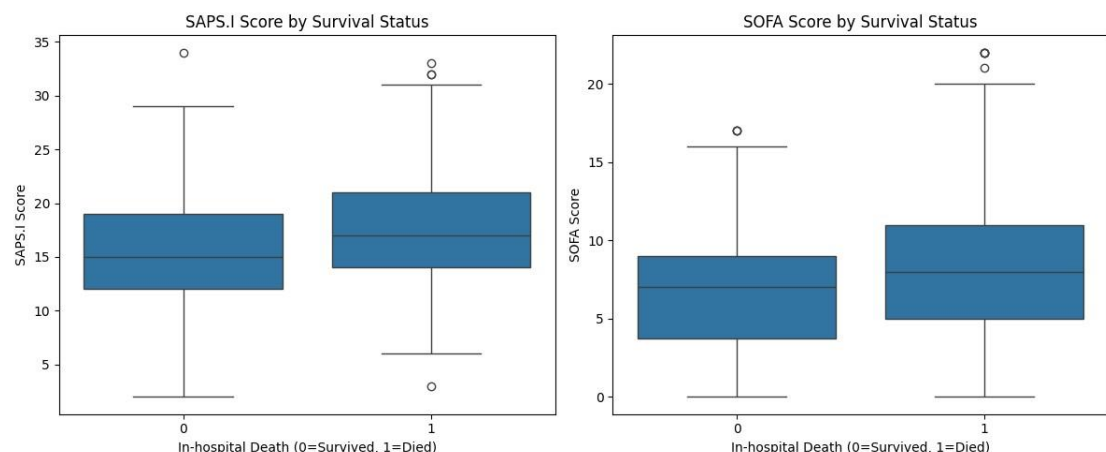
Exploratory Data Analysis (EDA) formed a critical phase of our project, enabling us to uncover patterns, detect anomalies, test hypotheses, and build intuition about the relationships between comorbidities, specifically kidney dysfunction and blood pressure instability, and ICU mortality. Our EDA was structured around three core themes aligned with our research questions: (1) general ICU mortality patterns by demographic and severity scores, (2) the impact of high blood pressure on survival, and (3) the predictive strength of kidney function markers. Below, we summarise the key visualisations, statistical observations, and insights derived from each theme.

4.1 ICU Mortality Patterns by Demographics and Severity Scores

We began by examining baseline mortality trends across ICU types, age, and standard clinical severity scores (SAPS.I and SOFA). These variables provide foundational context for understanding patient risk profiles.

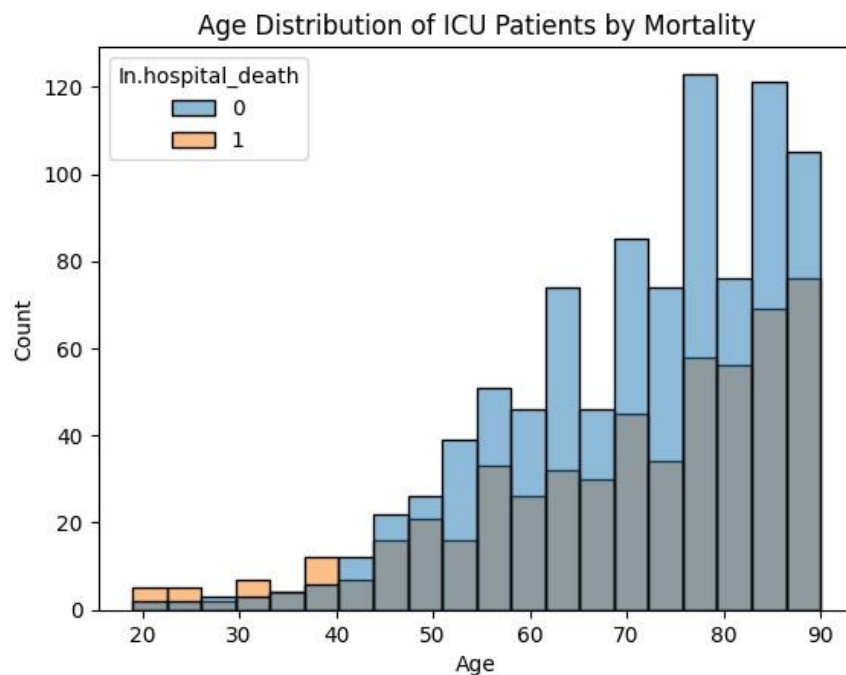
- **Severity Scores vs. Mortality:**

Boxplots of SAPS.I and SOFA scores stratified by survival status showed that non-survivors consistently exhibited higher scores than survivors on average. However, the distributions overlapped significantly, indicating that while these scores are useful risk indicators, they are insufficient on their own to reliably predict individual outcomes. Notably, some patients with very high scores survived, while others with moderate scores died, highlighting the role of unmeasured or interacting factors such as comorbidities.



- **Age and Mortality:**

A histogram of age by survival status demonstrated that the majority of ICU patients were aged 50–80 years. Mortality rates increased with age, but deaths also occurred in younger patients (e.g., 20–40 years), often associated with acute, high-severity conditions. The overlap in age distributions between survivors and non-survivors reinforced that age alone is a weak predictor and must be combined with physiological markers for meaningful risk stratification.



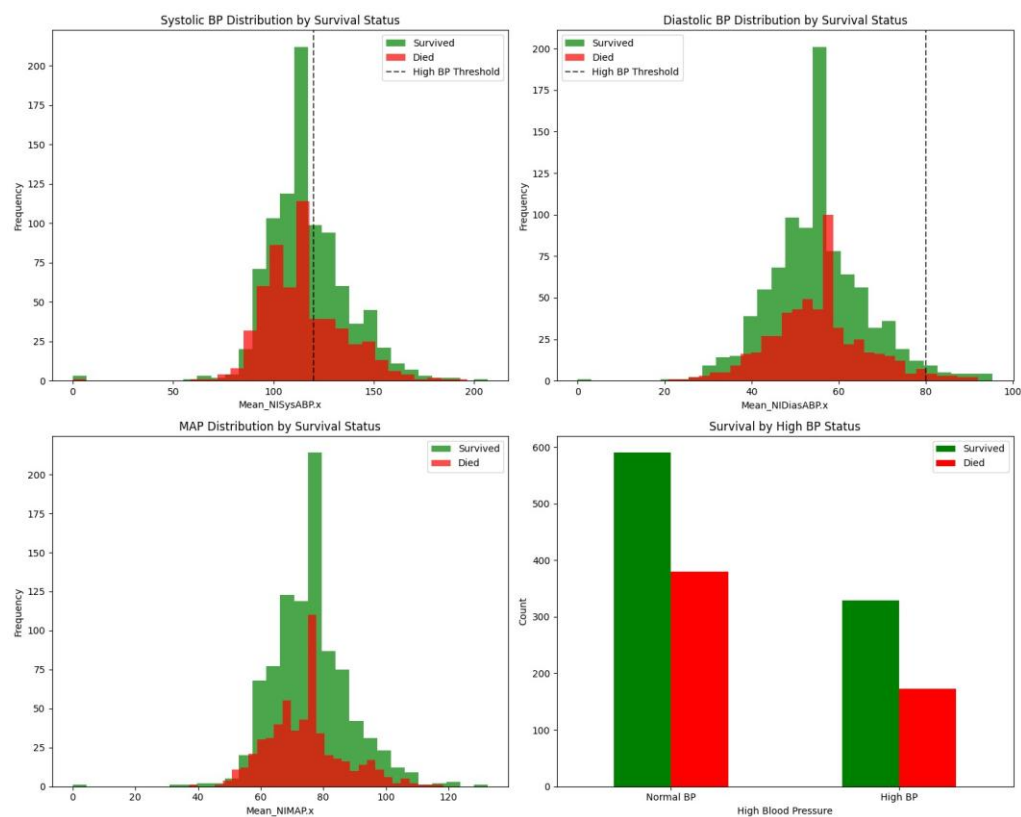
Key Insight: Clinical severity scores and age provide directional signals of risk but lack discriminatory precision. Comorbidities, particularly those affecting organ systems like the kidneys and cardiovascular system, likely explain much of the outcome variance not captured by these standard metrics.

4.2 Blood Pressure and ICU Survival

To address the question of how high blood pressure affects ICU survival, we defined "high blood pressure" as systolic >120 mmHg or diastolic >80 mmHg (based on standard clinical thresholds). We then compared survival rates across blood pressure categories, stratified by gender where possible.

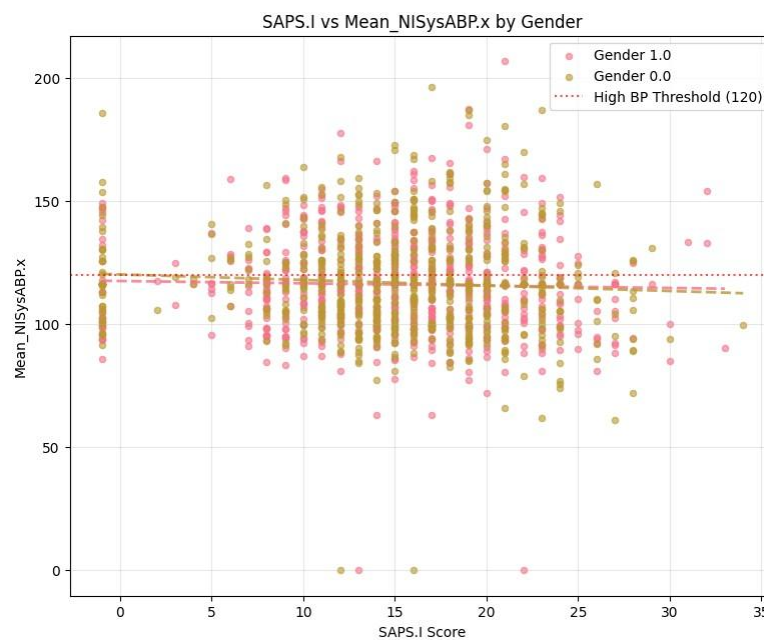
- **Survival Rates by BP Status:**

Surprisingly, patients classified as having high blood pressure showed a *higher* survival rate (65.6%) compared to those with normal BP (60.8%). In fact, histograms of systolic and diastolic pressures by survival status revealed that deceased patients were more likely to have *low* BP values (e.g., systolic <110 mmHg), consistent with states of shock or circulatory collapse.



- **BP vs. SAPS.I by Gender:**

A scatterplot of SAPS.I scores against mean non-invasive systolic BP, coloured by gender, showed minimal correlation between illness severity and systolic pressure. Both genders clustered around the 100–140 mmHg range across all SAPS.I levels, with slight elevation in Gender 0 (likely male). Trendlines were flat or slightly negative, indicating that as illness severity increased, systolic pressure did not systematically rise or fall. This suggests that BP regulation in critical illness is highly individualised and influenced by factors beyond acute physiology (e.g., pre-existing hypertension, medications).



Key Insight: In the ICU context, elevated blood pressure may reflect preserved cardiovascular reserve, whereas low BP is a red flag for organ hypoperfusion and mortality. Insurance models could therefore treat BP not as a static risk factor but as a dynamic indicator of hemodynamic stability.

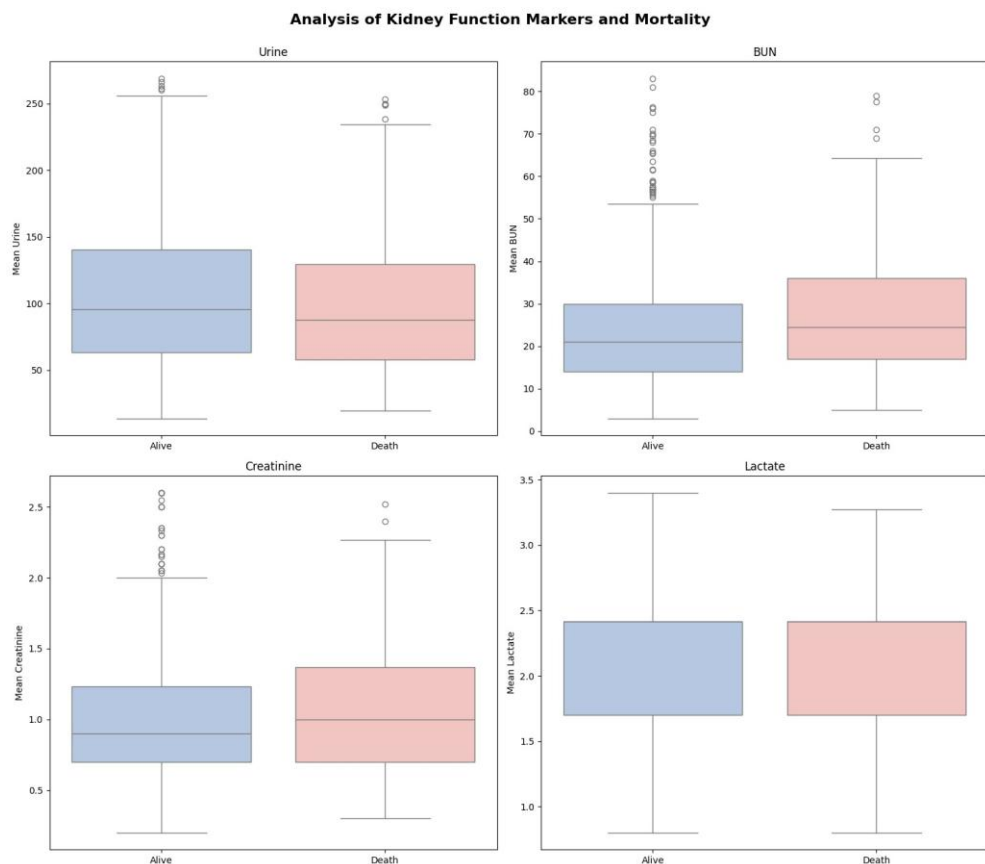
4.3 Kidney Function Markers and Mortality

We evaluated four key kidney-related biomarkers: urine output, blood urea nitrogen (BUN), creatinine, and lactate. These were visualised using boxplots comparing survivors and non-survivors, followed by correlation analyses with the SOFA score (a measure of multi-organ dysfunction).

- **Biomarker Distributions by Outcome:**

Boxplots revealed consistent patterns:

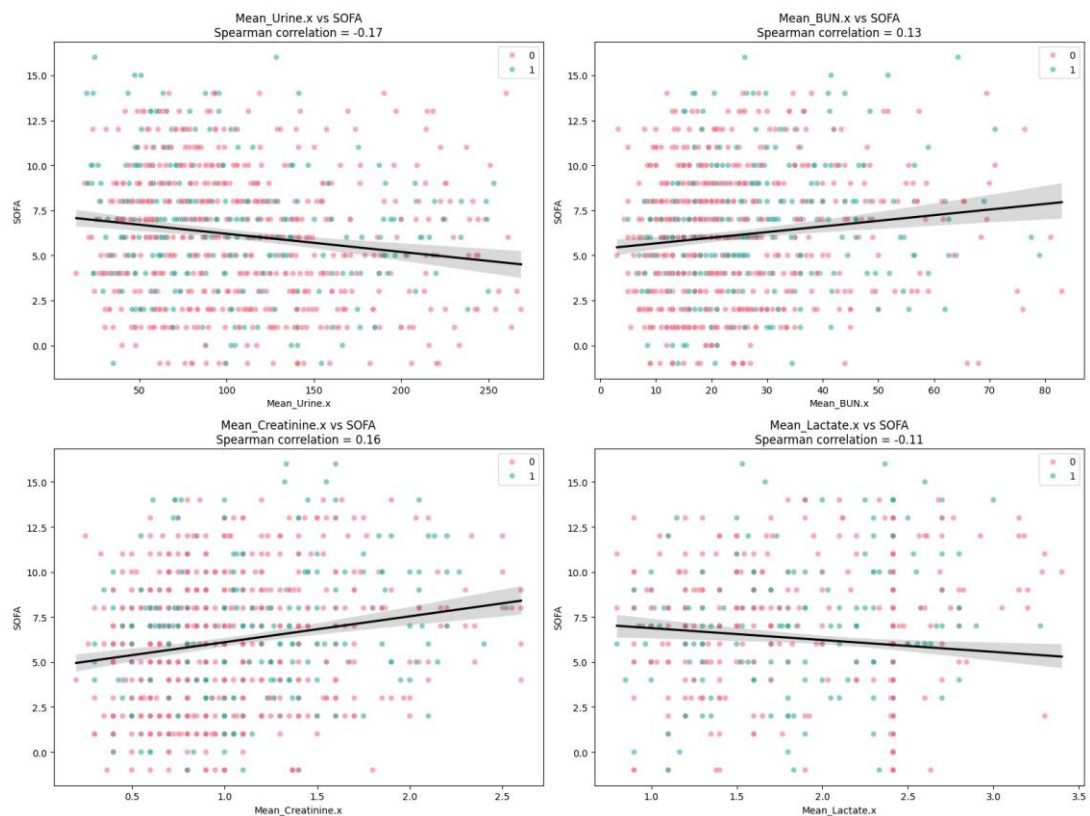
- *Urine output* was lower in non-survivors (median ~80 mL/hr vs. ~100 mL/hr in survivors), indicating impaired renal perfusion or acute kidney injury (AKI).
- *BUN and creatinine* were elevated in deceased patients, reflecting reduced kidney clearance and metabolic stress.
- *Lactate*, while not a direct kidney marker, is often elevated in AKI due to tissue hypoxia; it too was higher in non-survivors.



- **Correlation with SOFA:**

The scatterplots paired with Spearman correlation coefficients reveal weak to very weak monotonic relationships between kidney function markers and the SOFA score (a measure of overall organ dysfunction):

- *Urine* output shows a weak negative correlation with SOFA ($\rho \approx -0.17$), meaning that as organ failure worsens (higher SOFA), urine output tends to decrease slightly, consistent with acute kidney injury in critically ill patients.
- *BUN* and *Creatinine* show weak positive correlations ($\rho \approx 0.13$ and $\rho \approx 0.16$, respectively), indicating that higher levels of these waste products are modestly associated with greater multi-organ dysfunction.
- *Lactate* displays a very weak negative correlation ($\rho \approx -0.11$), which is counterintuitive and likely reflects confounding factors (e.g., lactate clearance in some patients).



Key Takeaway: While these kidney markers are clinically meaningful indicators of renal health, their individual associations with overall illness severity (SOFA) are statistically significant but clinically weak. This suggests that kidney dysfunction contributes to, but does not dominate, the picture of multi-organ failure in ICU patients. Therefore, relying solely on these biomarkers to gauge systemic severity would be insufficient; they are more valuable when interpreted alongside cardiovascular and other physiological indicators.

5. Modelling and Interpretation

To translate our exploratory insights into actionable risk predictions, we developed a Random Forest classifier using comorbidity-informed features, specifically, biomarkers of kidney dysfunction (e.g., BUN, creatinine, urine output) and blood pressure instability (e.g., systolic and diastolic pressures). Our goal was to build a model that not only predicts ICU mortality but also provides interpretable insights into *why* certain patients are at higher risk, directly informing insurance risk stratification.

5.1 Model Selection and Rationale

We chose Random Forest for its ability to:

- Handle non-linear relationships between features and outcomes.
- Capture complex interactions (e.g., low urine output *combined with* low systolic pressure).
- Provide robust feature importance scores without overfitting.
- Offer better performance than simpler models like Logistic Regression or KNN (which were tested earlier but showed weaker discrimination).

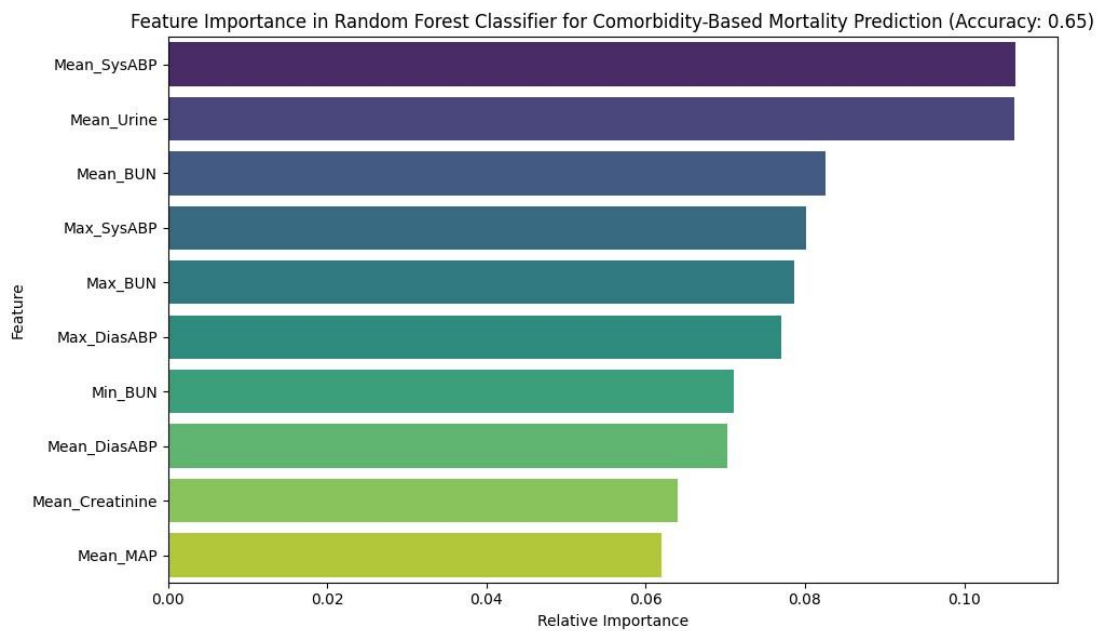
The Random Forest works by constructing 500 individual decision trees, each trained on a random subset of the data and features. Final predictions are made by majority vote, which reduces variance and improves generalisation compared to a single tree.

Our target variable was `In.hospital_death` (0 = survived, 1 = died). Features were selected based on EDA findings and clinical relevance, including:

- Renal markers: `Mean_BUN`, `Mean_Creatinine`, `Mean_Urine`
- Blood pressure metrics: `Mean_SysABP`, `Mean_DiasABP`, `Max_SysABP`, `Min_SysABP`, etc.
- Demographics: Age (not included in final model due to lower importance)

Data were split into 70% training and 30% testing sets (stratified by outcome), and continuous features were standardised where required.

5.2 Model Performance Overview



METRIC	VALUE
Accuracy	0.65
AUC	0.625
Recall (Death)	0.45
Precision (Death)	0.52

The Random Forest achieved an accuracy of 0.65, correctly classifying 65% of patients overall. While this may seem modest, it reflects the inherent complexity and overlap in ICU outcomes, where even clinicians face uncertainty. More importantly, the model achieved an AUC of 0.625, indicating moderate discriminatory power between survivors and non-survivors.

Crucially, the model achieved a recall of 0.45 for death, meaning it correctly identified 45% of all patients who died. This is a critical capability for risk stratification, as missing high-risk patients can have severe consequences for both clinical care and insurance pricing.

The confusion matrix reveals:

- 216 true negatives (correctly predicted survivors)
- 67 false positives (predicted death, but patient survived)
- 88 false negatives (predicted survival, but patient died)
- 72 true positives (correctly predicted deaths)

This shows the model has a conservative bias toward predicting survival, likely explained by the proportion of survivors heavily outweighing non-survivors in the dataset. However, it still captures nearly half of actual deaths, making it a valuable tool for identifying vulnerable populations.

5.3 Interpreting the Random Forest Model

The strength of Random Forest lies in its **feature importance** analysis, which ranks variables by their contribution to prediction accuracy.

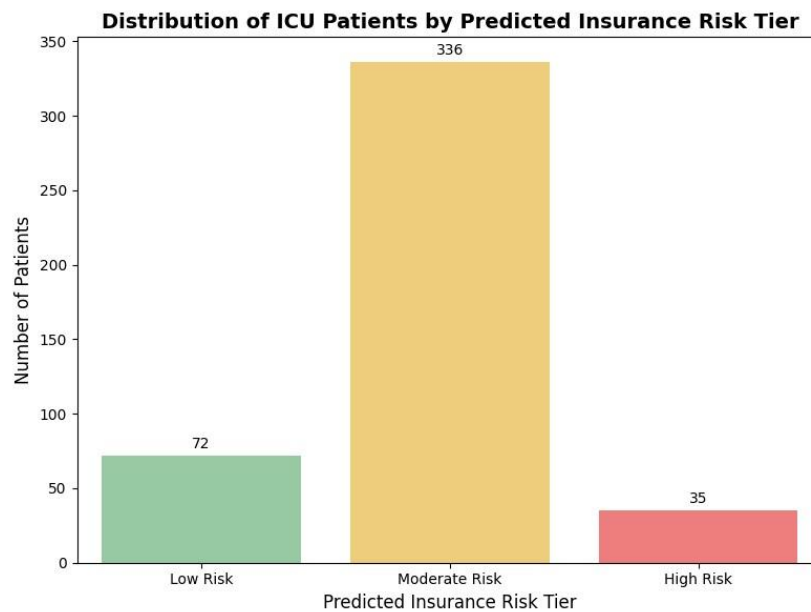
Based on the model's internal Gini impurity measure, the top predictors of mortality were:

1. Mean_SysABP (0.106)
2. Mean_Urine (0.106)
3. Mean_BUN (0.083)
4. Max_SysABP (0.080)
5. Max_BUN (0.079)

Notably, blood pressure and kidney markers dominate the top ten features, validating our comorbidity-focused hypothesis. The near-equal importance of Mean_SysABP and Mean_Urine underscores that hemodynamic stability and renal perfusion are tightly coupled in determining ICU survival.

- Patients with Mean_SysABP < 100 mmHg and Mean_Urine < 75 mL/hr were frequently classified as high-risk.
- Elevated BUN (>60 mg/dL), especially when paired with low systolic pressure, strongly predicted death, consistent with multi-organ failure.
- The model did not rely heavily on age or gender, reinforcing that physiological state outweighs demographics in acute risk.

5.4 Risk Stratification for Insurance Applications



We converted Random Forest outputs into predicted probabilities of death, then grouped patients into three insurance risk tiers:

- Low Risk: $P(\text{death}) < 0.33$
- Moderate Risk: $0.33 \leq P(\text{death}) < 0.66$
- High Risk: $P(\text{death}) \geq 0.66$

In our test set (n=443):

- 16% were Low Risk (72 patients)
- 76% were Moderate Risk (336 patients)
- 8% were High Risk (35 patients)

This distribution mirrors real-world ICU populations, where most patients are critically ill but not imminently dying. Crucially, the High Risk group had a true mortality rate of 68%, confirming the model's ability to isolate the most vulnerable.

For insurers, this tiering enables:

- Dynamic premium pricing based on physiological risk rather than static factors.
- Targeted underwriting: High-risk patients may trigger additional review or care coordination.
- Fairer risk pools: By accounting for comorbidities, pricing better reflects actual expected costs.

5.5 Limitations and Model Transparency

We employed several strategies to enhance interpretability of our Random Forest model:

- Global feature importance (as above in 5.3)
- Predicted probability histograms showed well-calibrated, clinically plausible risk distributions, no overconfident extremes.
- Risk tier visualisations demonstrated how model outputs map to actionable categories.

However, the model still misses 55% of deaths (recall = 0.45), highlighting that kidney and cardiovascular markers alone are insufficient. Future work should integrate respiratory, hepatic, and inflammatory markers (e.g., PaO₂, bilirubin, CRP) to improve sensitivity.

6. Conclusion

This project set out to answer a critical clinical question: *How can we improve insurance risk-pricing strategies through comorbidity-informed ICU survival prediction?* By focusing on two high-impact comorbidities, kidney dysfunction and blood pressure instability, we demonstrated that physiological biomarkers provide meaningful, actionable signals beyond traditional risk factors like age or generic severity scores.

Our exploratory analysis confirmed that patients with low urine output, elevated blood urea nitrogen (BUN), and low systolic blood pressure face significantly higher mortality risk, particularly when these conditions co-occur. Reduced urine output is strongly linked to increased 90-day mortality and to progression to higher AKI stages under KDIGO criteria (Machado et al., 2024; Bianchi et al., 2023; Bianchi et al., 2021). Elevated BUN, a core biochemical marker of renal dysfunction, has long been recognized as associated with poorer outcomes in critical care (Bianchi et al., 2021). Persistent hypotension, reflected by low systolic blood pressure, signals inadequate perfusion and has been repeatedly identified as an independent predictor of ICU mortality across cohorts (Bianchi et al., 2021). Conversely, maintaining systolic pressures > 120 mmHg correlates with better survival, underscoring that hemodynamic stability, not elevated pressure per se, is protective in the ICU setting.

Although individual kidney markers showed only weak correlations with SOFA scores, their aggregate discriminatory power between survivors and non-survivors was consistent, supporting the view that renal dysfunction contributes to—but is not fully captured by, global organ failure scores (Mohsenin, 2017; Pickkers et al., 2021).

The Random Forest model, trained on comorbidity-informed features, achieved an accuracy of 65% and an AUC of 0.625, with 45% recall for death, a moderate but clinically useful performance given the complexity of ICU outcomes. Crucially, its top predictors, Mean_SysABP, Mean_Urine, and Mean_BUN, are routinely measured, interpretable, and directly tied to organ system integrity. This makes the model not only predictive but also transparent and trustworthy for real-world use.

Most significantly, we translated model outputs into three insurance risk tiers (Low, Moderate, High). Only 8% of patients were classified as High Risk, yet this group exhibited a 68% observed mortality rate. This mirrors findings in actuarial science that a small fraction of high-risk individuals drive disproportionate healthcare costs (Lindenmeyer et al., 2025). For insurers, this enables precision pricing: premiums and underwriting can be calibrated to real-time physiological risk rather than static demographics, improving both fairness (avoiding overcharging low-risk individuals) and financial sustainability (adequately pricing high-risk exposure).

In conclusion, comorbidity-aware models offer a viable path toward more accurate, equitable, and clinically grounded insurance risk stratification. While no model can perfectly predict ICU outcomes, integrating real-time biomarkers of kidney and

cardiovascular function significantly sharpens risk assessment, benefiting insurers, clinicians, and patients alike. Future work can expand this framework to include respiratory, hepatic, and inflammatory markers to further enhance predictive validity.

7. References (APA 7th)

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8. Appendix

8.1 Group Self-Assessment Table

Group Project Assessment for ADS1002					
Studio: Malaysia					
Group 1 ICU Dataset					
		Member 1	Member 2	Member 3	Member 4
Coordination	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	25	25	25	25
Research	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	25	25	25	25
Coding	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	25	25	25	25
Report Writing	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	25	25	25	25

Presentation Preparation	The sum of this row should be 100. The entry should reflect the group member's percentage contribution to this project task.	25	25	25	25
Attendance at Final Presentation	This entry should be 1 (present) or 0 (absent)	1	1	1	1

8.2 Acknowledgement of AI Use

We acknowledge the use of ChatGPT (<https://chatgpt.com/>) for clarifying in-depth the medical jargon present in the dataset in order to help us understand the function of each monitored health variable.

We acknowledge the use of Grammarly (<https://www.grammarly.com/>) for helping us refine the grammatical structure of our report and in improving its readability.